

## motion in EM fields

The direction of the **magnetic force** is at **right angles** to the plane containing the magnetic field and to the **velocity of a moving charge**.

The force itself is **proportional** to:

- the velocity of the charge,  $v$
- the magnitude of the charge,  $q$
- the magnetic field strength,  $B$

The **magnitude** of magnetic force  $F$  is given by  $F = qvB \sin(\theta)$  where  $\theta$  is the angle between the **velocity of the charge** and the **magnetic field**.

$$\therefore F = qvB \implies \sin(\theta) = 1 \implies \theta = 90^\circ.$$

**Magnetic field strength**  $B$  is measured in tesla (T).

There is another way to express  $F$ ; in a conductor that carries an electric current  $I$  of charge carriers.

$F = qvB \sin(\theta) \equiv BIL \sin(\theta)$ , where  $\theta$  would be defined as the angle between the magnetic field and the direction of the current.  $L$  is the length of the conductor.

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Regarding **Fleming's left-hand rule**,

Thumb: Force

Index finger: Magnetic field

Middle finger: Conventional current

Remember: **F**ather, **M**other **C**hild

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⊙ illustrates a current, or field direction, coming out of the paper

⊗ illustrates a current, or field direction, going into the paper

Magnetic force  $F$  acting on one wire (call it wire  $\alpha$ ) due to the magnetic field of another wire (call it wire  $\beta$ ) must be:

$$F = (\text{magnetic field due to wire } \beta \text{ at the position of wire } \alpha) \times I_{\alpha} \times L_{\alpha}$$

Therefore, the **force per unit length** acting on wire  $\alpha$  is:

$$\frac{F}{L_{\alpha}} = B \times I_{\alpha}$$

The magnetic field due to a current at a perpendicular distance  $r$  from a wire carrying current  $I$  is given by:  $B = \mu_0 \frac{I}{2\pi r}$ .

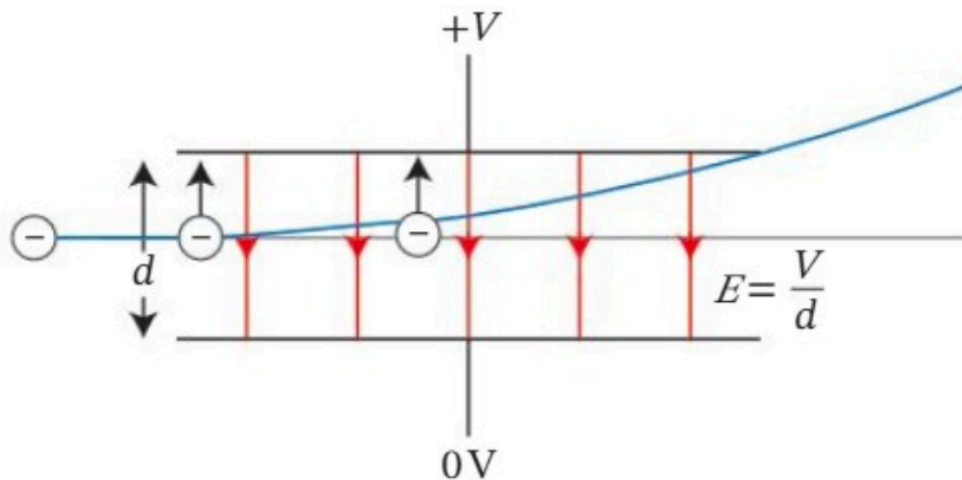
Hence, for the case of wires  $\alpha$  and  $\beta$  here, the force per unit length acting on wire  $\alpha$  due to the magnetic field of wire  $\beta$  is:

$$\begin{aligned} \frac{F}{L_{\alpha}} &= \left( \mu_0 \frac{I_{\alpha}}{2\pi r} \right) \times I_{\beta} \\ &= \mu_0 \frac{I_{\alpha} I_{\beta}}{2\pi r} \end{aligned}$$

*note that this is not substituting  $B$  in the previous equation!*

The expression above is **symmetrical**, showing that the magnitude of the force of wire  $\beta$  on wire  $\alpha$  is equal to that of wire  $\alpha$  on wire  $\beta$ . This is supported by Newton's third law.

Regarding charged particles in a uniform **electric field**,



Since  $F = q \times E$ , the force on the charge is  $Ee$  and the acceleration is  $a = \frac{Ee}{m_e}$  because  $F = ma$ ;  $m_e$  is the mass of the electron.

The **electric field**  $E$  is produced by the **potential** difference  $V$  between the plates, which are separated by a distance  $d$ , so  $E = \frac{V}{d}$ .

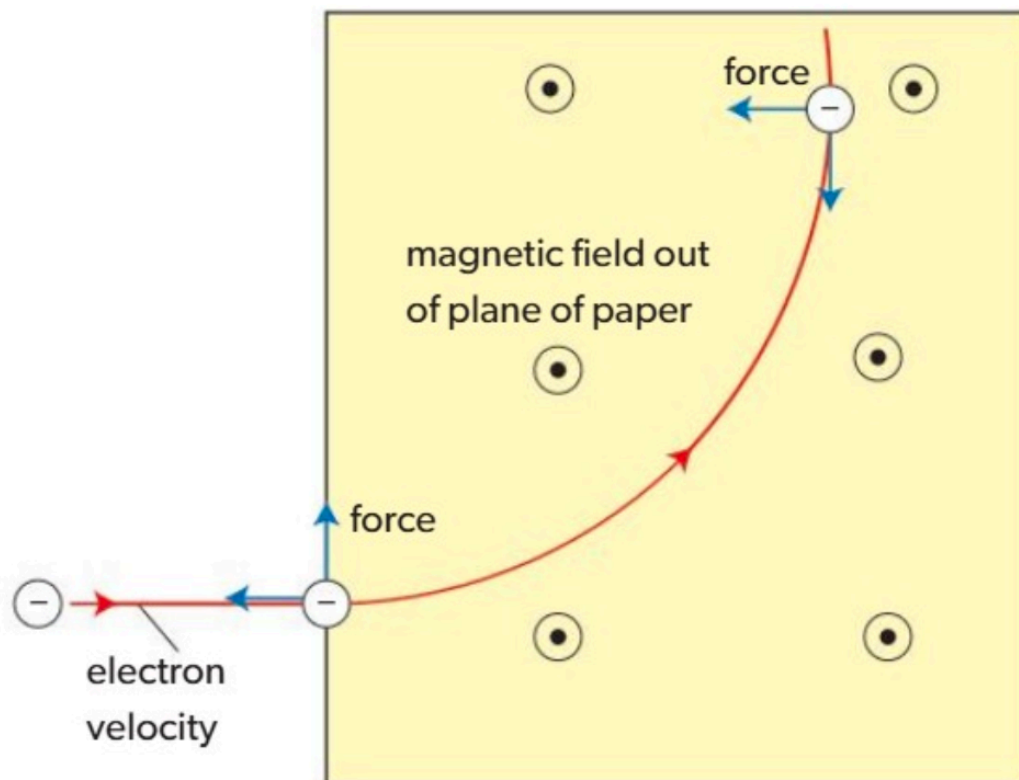
And so,

$$\begin{aligned}a &= \frac{Ee}{m_e} \\ E &= \frac{V}{d} \\ \therefore a &= \frac{V}{d} \times \frac{e}{m_e} \\ &= \frac{eV}{m_e d}\end{aligned}$$

*This is not found in the formula booklet so good luck rofl xD*

In any practical situation, weight can be ignored because the electric force  $\gg$  weight.

*In my nightmares*,  $\exists$  a charge particle of mass  $m$  that enters a uniform magnetic field of strength  $B$  perpendicular to the field lines. The charge on the particle is  $q$ .



▲ **Figure 5** The electron is moving perpendicular to a magnetic field. The force on the electron is at right angles to its motion so the magnetic force is providing a centripetal force that acts on the electron

The magnetic force always acts at  $90^\circ$  to the velocity of the particle. **This is the condition for a centripetal force**, so the particle moves in a circular path. *Insert lightning strike. DUUN DUN DUUUUUUN sound effect.* The speed of the particle is invariant.

Hence, this (kind of) relates to [orbits in gravitational fields](#).

The force on the particle is  $qvB$ , which must equal  $\frac{mv^2}{r}$ , where  $r$  is the radius of the orbit. The **radius of the orbit** can be expressed in several ways:

$$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mE_k}}{qB}$$

The derivation for  $r = \frac{mv}{qB}$  is as follows:

$$\begin{aligned} qvB &= \frac{mv^2}{r} \\ qB &= \frac{mv}{r} \\ r \times qB &= mv \\ r &= \frac{mv}{qB} \\ &\text{QED} // \end{aligned}$$

The derivation for  $r = \frac{p}{qB}$  is as follows:

$$\begin{aligned} qvB &= \frac{mv^2}{r} \\ r &= \frac{mv}{qB} \quad (\text{as shown previously}) \\ p &= mv \\ \therefore r &= \frac{p}{mv} \\ &\text{QED} // \end{aligned}$$

Note that  $p$  is the momentum of the charged particle.

The derivation for  $r = \frac{\sqrt{2mE_k}}{qB}$  is as follows:

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv^2}{qvB}$$

$$r = \frac{pv}{qvB}$$

$$r = \frac{p}{qB}$$

$$E_k = \frac{p^2}{2m} \quad (\text{according to formula bookie})$$

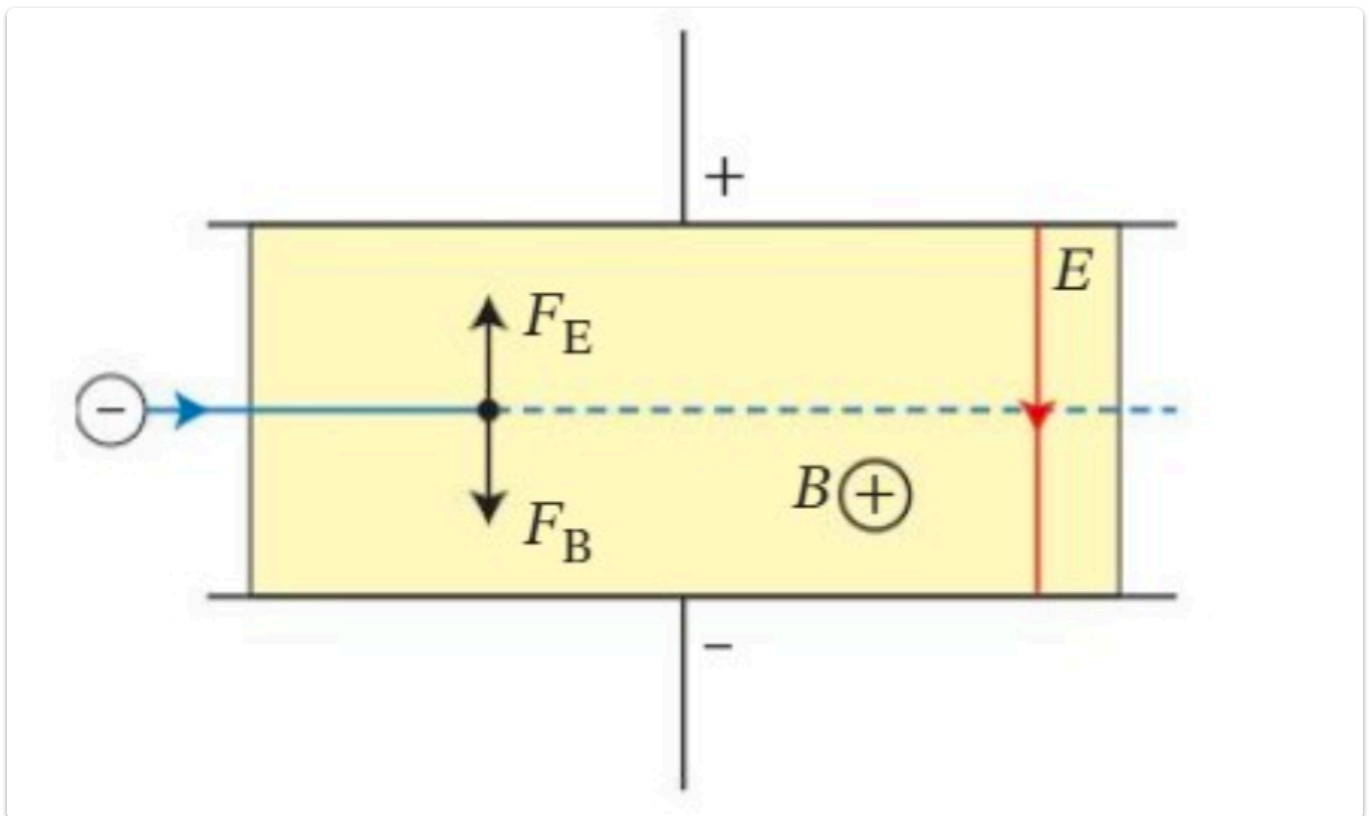
$$p^2 = E_k \times 2m$$

$$p = \sqrt{2mE_k}$$

$$\therefore r = \frac{\sqrt{2mE_k}}{qB} \quad (\text{through substitution})$$

*QED//*

Consider the movement of [charged particles](#) in perpendicular electric and magnetic fields. An example is shown below.



The force as a result of the [electric field](#)  $E$  is  $F_E$ . It's direction is due to the fact that like charges repel and opposing charges attract.

Using [Fleming's left-hand rule](#), the force that acts on the charge as a result of the magnetic field  $B$ ,  $F_B$  has a direction going downwards.

In the example image above,  $F_E$  and  $F_B$  act on the charged particle with the same magnitude, where there will be no resultant force; this assumes that the gravitational force is negligible.

And so, for this condition,

$$\begin{aligned}F_E &= F_B \\qE &= qvB \\E &= vB \\v &= \frac{E}{B}\end{aligned}$$

$v$  would be the single speed at which the particle will travel **without deflection**. At all other speeds, the particle will be deflected upwards (at speeds less than  $v \implies F_E > F_B$ ) or downwards (at speeds greater than  $v \implies F_E < F_B$ ).

This is useful as a velocity **filter** to select specific particles for an experiment.